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Lab session # 4
ALTEGRAD 2023

11/11/26

1 Question 1

G is the disjoint union of K_{100} and $K_{50,50}$. Since the two components share no vertex and no edge joins them, edges and triangles can be counted separately in each component and summed.

Number of edges. In the complete graph K_n, every unordered pair of distinct vertices is joined by exactly one edge, so $|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2}$. In the complete bipartite graph $K_{a,b}$, every vertex of one part is joined to every vertex of the other part and no edge lies within a part, so $|E(K_{a,b})| = ab$. Hence

$$|E(G)| = \binom{100}{2} + 50 \times 50 = 4950 + 2500 = 7450. \tag{1}$$

Number of triangles. A triangle is a set of three pairwise adjacent vertices. It is connected, so it lies entirely within one connected component.

- In K_{100}, any three distinct vertices are pairwise adjacent, so every 3-subset of vertices forms a triangle. This gives $\binom{100}{3} = \frac{100 \cdot 99 \cdot 98}{6} = 161700$ triangles.
- In $K_{50,50}$, any three vertices are distributed over only two parts, so by the pigeonhole principle at least two of them lie in the same part. These two are not adjacent, so no three vertices are pairwise adjacent and there are 0 triangles. Equivalently, a bipartite graph contains no odd cycle, and a triangle is a cycle of length 3.

Hence

$$t(G) = \binom{100}{3} + 0 = 161700. \tag{2}$$

2 Question 2

Smallest eigenvalue and its eigenvector. Let $\mathbf{1} \in \mathbb{R}^m$ be the all-ones vector. Each row of $D^{-1}A$ sums to $\frac{1}{d_i}\sum_j A_{ij} = 1$, so $D^{-1}A\mathbf{1} = \mathbf{1}$ and

$$L_{rw}\mathbf{1} = \mathbf{1} - D^{-1}A\mathbf{1} = 0. \tag{3}$$

Hence $\mathbf{1}$ is an eigenvector of L_{rw} with eigenvalue 0. This is the smallest eigenvalue. Indeed, $L_{rw} = D^{-1/2}L_{sym}D^{1/2}$ with $L_{sym} = D^{-1/2}(D - A)D^{-1/2}$ symmetric, so the eigenvalues of L_{rw} are real, and $L_{rw}u = \lambda u$ is equivalent to $(D - A)u = \lambda Du$. Multiplying by $u^\top$ gives

$$\lambda\, u^\top Du = u^\top(D - A)u = \frac{1}{2}\sum_{i,j} A_{ij}(u_i - u_j)^2 \geq 0, \tag{4}$$

and $u^\top Du > 0$ since every node of a connected graph (with at least two nodes) has positive degree, so $\lambda \geq 0$. Moreover, $\lambda = 0$ if and only if $u_i = u_j$ for every edge (i, j). Since the graph is connected, any two nodes are joined by a path, so u is constant on all nodes. The eigenvalue 0 is therefore simple, and the corresponding eigenvector is the constant vector $u_1 = c\,\mathbf{1}$, $c \neq 0$.
Intuitively, the eigenvectors of eigenvalue 0 encode the connected components of the graph: with p components, the indicator vectors of the components span this eigenspace. For a connected graph there is a single component, so u_1 describes the trivial partition into one cluster and maps every node to the same value.

Effect of ignoring this eigenvector. The first column of U adds the same coordinate c to every row y_i, so it does not change any pairwise distance: $\|y_i - y_j\|^2$ is identical with or without it. The k-means objective, the sum of squared distances to the cluster means, is invariant under such a translation, so k-means returns the same partition. Ignoring u_1 therefore has *no effect* on the clustering: it carries no information that can separate nodes. If instead it is replaced by the next eigenvector u_{d+1}, keeping d dimensions, the embedding gains an informative direction, which can change, and typically improve, the clustering.

3 Question 3

4 Question 4

5 Question 5